Introduction to probability, statistics and data handling

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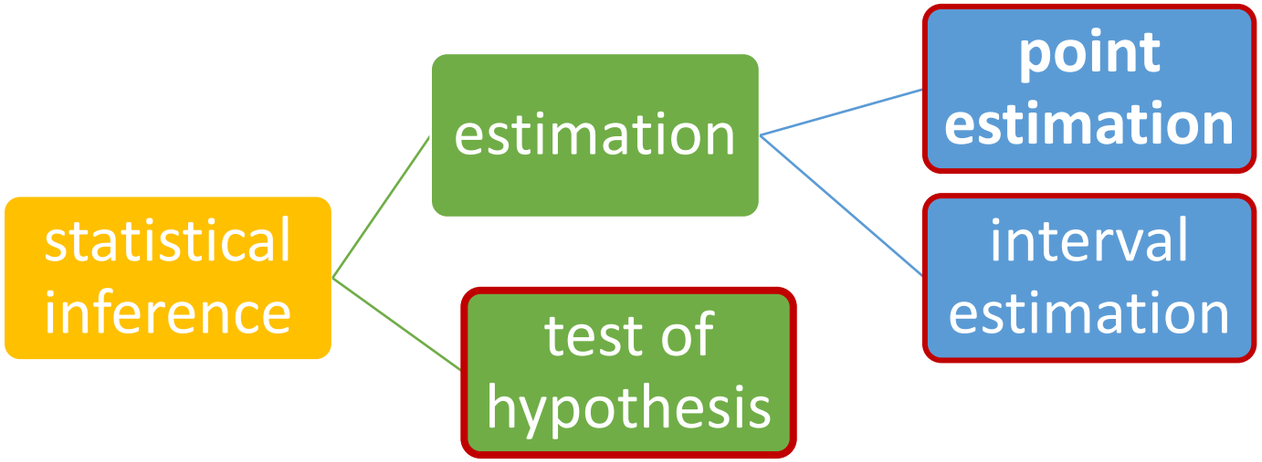
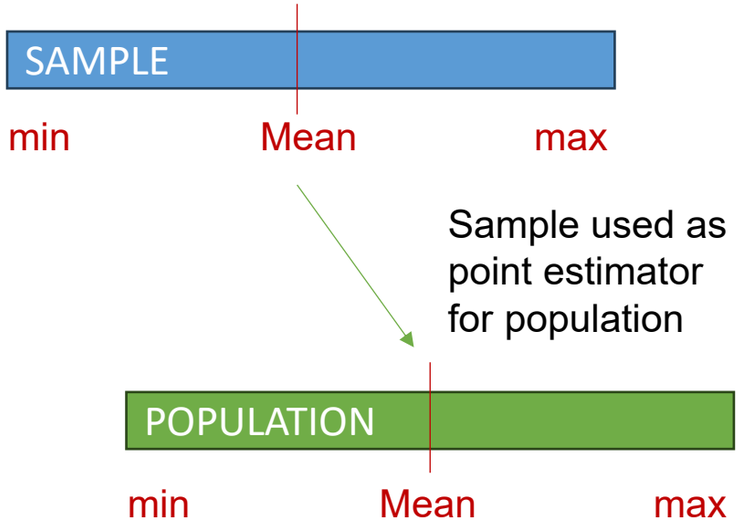
Statistical Inference

- The statistical inference consists in arriving at (quantitative) conclusions concerning a **population** where it is impossible or impractical to examine the entire set of observations that make up the population. Instead, we depend on a **subset** of observations - a **sample**.
- This process of analysing sample data and making conclusions about the parameters of a population is called **Statistical Inference**.





3 Forms of Statistical Inference

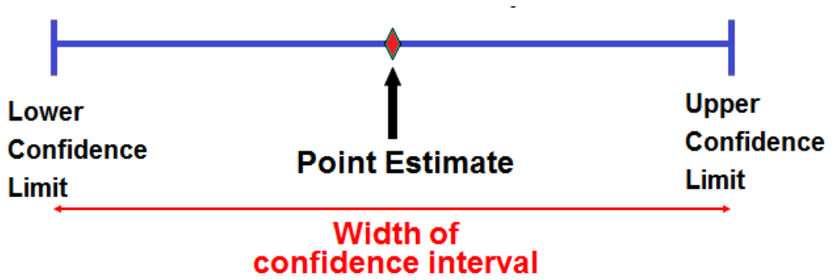
Statistical Inference

- This process of analysing sample data and making conclusions about the parameters of a population is called **Statistical Inference**.

Ex. – point estimation:
I checked 5 restaurants in Milano and claim that mean cost of pizza is 15 €.

Ex. interval estimation:
In Milano pizza costs between 10 and 20 €.

Ex. hypothesis testing:
I bought a pizza in Milano for 18 €. Verify if the cost of pizza in Milano is higher than 15 €.



Do I always need to repeat the experiment to obtain the result?

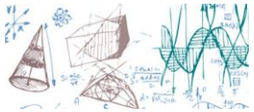
Point estimation

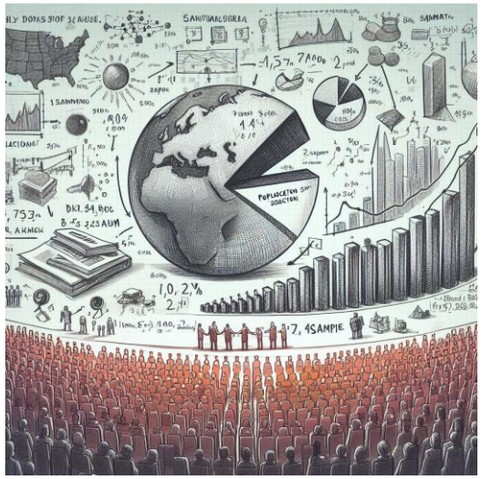
- **Point estimation** in statistics involves using sample data to calculate a single value that serves as the **best approximation** of an unknown population parameter.

Suppose we want to estimate the **mean weight** of a certain species of turtles in Florida. Collecting data on every individual turtle in Florida would be impractical due to the large population size. Instead, we take a **random sample** of 50 turtles and use the **sample mean weight** to estimate the true population mean.

- **Population Parameter**: The true average weight of all turtles in Florida (population mean, denoted as μ).
- **Sample Data**: We weigh 50 turtles and find the **sample mean weight** to be **150.4 pounds**.
- Our **point estimate** for the true population mean weight is **150.4 pounds**. This value represents our best guess based on the sample data.

Statistical Sample and Population

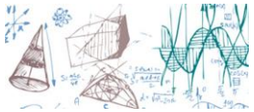




- We start with three estimators:
  - estimator of a **mean value**
  - estimator of a **variance**
  - estimator of **proportion**

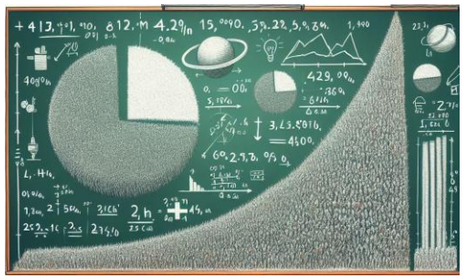
  we want to estimate μ, σ^2 and p of a **population** with a use of **sample**

- It would be perfect if the estimator value of the parameter and the value of parameter were equal.
- It would be nice if the poin estimator went closer and closer to the parameter if we, for example, increase the sample size
- Later we will develop methods for the estimation of unknown parameter of a model (linear, or any other) based on samples (method of moments, method of least squares, maximum likelihood estimation)



Point estimation

- Let’s think about the following: we are looking at some phenomena (took a data sample), now what we like to do is to try describe the data using a model (have we already discussed any models?)
- Using the statistics lingo we would say: we want to estimate the parameters for the hypothesised population model
- **As usual there are a lot of methods, we are going to have a look at a few of them**
- Estimators should have specific features (we will discuss it today)
  - **BUT**
- Let’s start with some **examples** first!



Statistical Sample and Population

- Sample posses a property X (our RV); $X \rightarrow f(x, \boldsymbol{\lambda})$ (probability density function), $\boldsymbol{\lambda}$ – set of parameters of the population to be determined from the sample (e.g. μ, σ, etc.).
- Any function of the random variables constituting a random sample that is used for **estimation** of unknown distribution parameters $\boldsymbol{\lambda}$ is called a **statistic** $\boldsymbol{S}$:

$$S = S(X_1, X_2, \dots, X_n)$$

$$\lambda_i = E[S(X_1, X_2, \dots, X_n)] \equiv \hat{S}$$

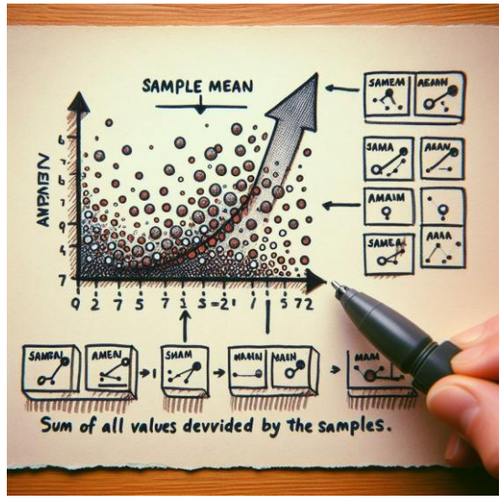
- We say: the estimated value of a statistic $\hat{S}$ is said to be estimator of the parameter λ ; the estimation is carried out on the basis of an n-element **sample**.

do we know any statistic?

Estimators

- Let's set a **generic procedure** using this simple example
- First, we **pick the parameter** to be estimated
- Next, we need to **collect data** and **compute a sampling statistics** using a formula corresponding to the parameter we are interested in
- In our example that is a **sample mean**

$$\boxed{\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}}$$



- This, in turn, we call an **estimator** of true parameter, in our case this would be: $\mu \rightarrow \bar{X} = \hat{\mu}$ (we use the caret symbor "^")
- Remember – the estimator is a random variable, for different sample we are going to get different value
- The estimator will follow its own distribution – **sampling distribution of the estimator**

Parameter estimation

- A **point estimator** is a function of observed data used to estimate an unknown population parameter.
- If we take a random sample: $X_1, X_2, .., X_n$,

  which come from a distribution $f(x;\theta)$,

  then a point estimator is a parameter based on random sample: $\hat{\theta} = T(X_1, X_2, .., X_n)$

  - Therefore, the main concern is to find a model, i.e. $f(x;\theta)$.
  - We can start with the most well-known.

Parameter estimation

The parameters of a pdf are any constants that characterize it,

$$f(x;\theta) = \frac{1}{\theta} e^{-x/\theta}$$

r.v. parameter

i.e., θ indexes a set of hypotheses.

Suppose we have a sample of observed values: $\boldsymbol{x} = (x_1, \ldots, x_n)$

We want to find some function of the data to estimate the parameter(s):

$$\boxed{\hat{\theta}(\vec{x})}$$

← estimator written with a hat

Sometimes we say 'estimator' for the function of $x_1, \ldots, x_n$; 'estimate' for the value of the estimator with a particular data set.

Estimators

- Consider the following: to check the water for contamination by a micro-organism a number of samples were taken, the results are summarised as follow

Counts	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**>9**
Frequency	53	25	13	2	2	1	1	0	1	0

- One can assume that the data follow the Poisson distribution with an unknown parameter μ (each water sample is an independent observation on the same random variable!)
- For these particular data, we can estimate the μ as:

$$\bar{x} = \frac{0 \cdot 53 + 1 \cdot 25 + \cdots + 8 \cdot 1}{58 + 25 + \cdots + 1} = \frac{84}{103} = 0.816$$

$$\{X_1, X_2, \ldots, X_{103}\} \rightarrow X \equiv Poisson(\mu)$$

$$\bar{X}_{(1)} = \frac{X_1 + X_2 + \cdots + X_{103}}{103} \rightarrow \qquad \bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

Properties of estimators

1. Bias

The bias of a point estimator is defined as the difference between the expected value of the estimator and the value of the parameter being estimated. When the estimated value of the parameter and the value of the parameter being estimated are equal, the estimator is considered unbiased.
Also, the closer the expected value of a parameter is to the value of the parameter being measured, the lesser the bias is.

2. Consistency

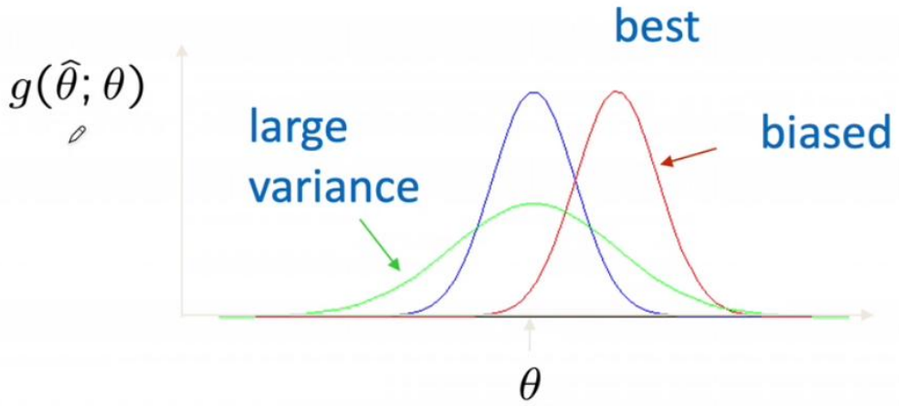
Consistency tells us how close the point estimator stays to the value of the parameter as it increases in size. The point estimator requires a large sample size for it to be more consistent and accurate. You can also check if a point estimator is consistent by looking at its corresponding expected value and variance. For the point estimator to be consistent, the expected value should move toward the true value of the parameter.

3. Most efficient or unbiased

The most efficient point estimator is the one with the smallest variance of all the unbiased and consistent estimators. The variance measures the level of dispersion from the estimate, and the smallest variance should vary the least from one sample to the other.
Generally, the efficiency of the estimator depends on the distribution of the population. For example, in a normal distribution, the mean is considered more efficient than the median, but the same does not apply in asymmetrical distributions.

Properties of estimators

If we were to repeat the entire measurement, the estimates from each would follow a pdf:

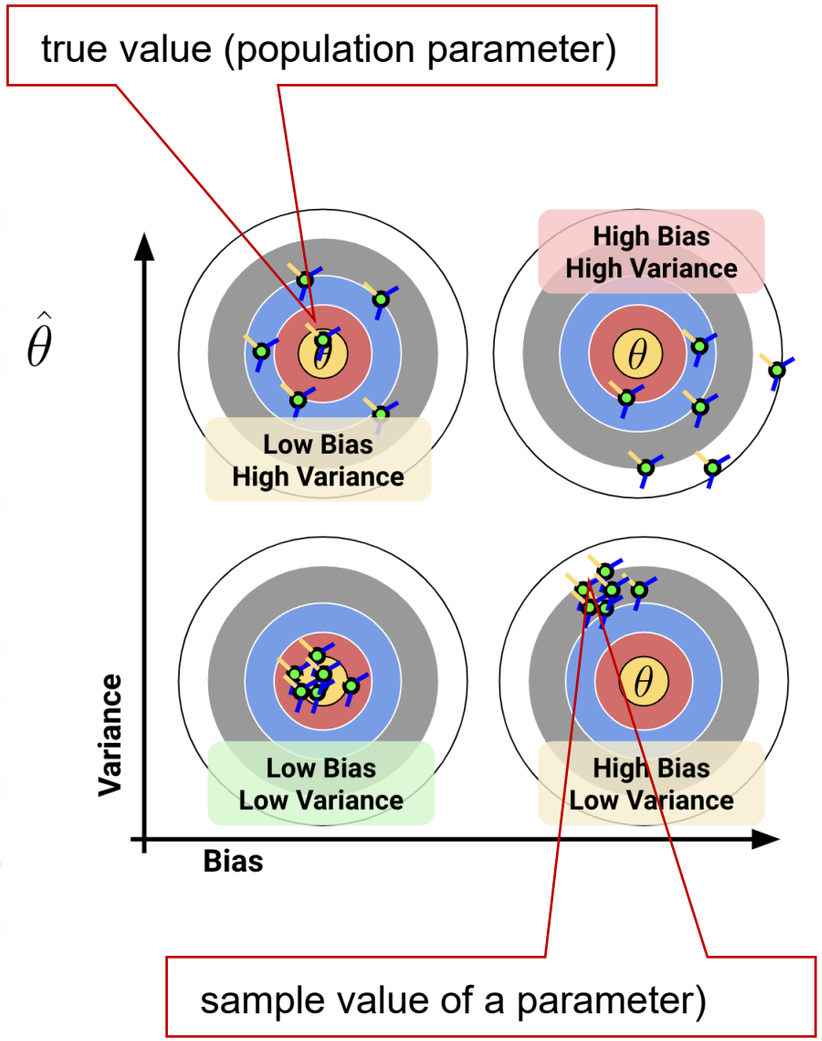


We want small (or zero) bias (systematic error): $b = E[\hat{\theta}] - \theta$

→ average of repeated measurements should tend to true value.

And we want a small variance (statistical error): $V[\hat{\theta}]$

→ small bias & variance are in general conflicting criteria



Estmator Wish List

- We are looking for the best estimator (but what does „best” mean?
- In the best of all possible worlds, we could find an estimator $\hat{\mu}$ for which $\hat{\mu} = \mu$ in all samples. But this does not exist, sometimes $\hat{\mu}$ will be too small, fort other samples too big.
- Let’s write (in general): $\hat{\theta} = \theta + error\ of\ estmation$. Therefore, the best estimator $\hat{\theta}$:
  - has small estimator errors: the mean squared error RMS $E\left[\left(\hat{\theta} - \theta\right)^2\right]$ should be the smallest
  - should be **unbiased** $E\left[(\hat{\theta})\right] = \theta$
  - should have small variance $VAR\left[(\hat{\theta})\right]$
- We are looking for **unbiased** (expectation value) and **efficient** estimators (variance).

Summary so far

- A generic „algorithm” for point estimation task would be:
- **Collect the data** and understand it
- Come up with a **model**, this will specify a **parameter** or many parameters that we need to make an estimate
- For a given parameter(s) we need an **estimator(s)** (typically we will concentrate on the mean value or variance, however we also can tackle more ambitious cases – e.g., divorces)
- Work out the **estimate of the parameter** – this is a random variable and will be different for different data sets
- Finally, analyse the **sampling distribution of the estimator** to make a judgement of its usefulness
- We are looking for **unbiased** (expectation value) and **efficient** estimators (variance)

Sampling distribution

- Any sample statistics is a function of R.Vs and is therefore itself a random variable – that is absolutely critical to remember!
- The probability distribution of a sample statistics is called **the sampling distribution** of this statistics (sorry for complicated circular sentences…)
  - A recipe to get such distribution would be as follow: we should draw all possible samples of size n from a population, next we should compute the statistics at hand, thus, obtaining the distribution of this statistics. We call it the **sampling distribution**
- It is perfectly ok to compute the mean, variance, standard deviation and other moments for the sampling distribution!
- To make it a bit more comprehensible, let's consider the sample mean. Let $X_1, X_2, \cdots, X_n$ be independent, identically distributed RVs. The mean of the sample is another R.V. defined as follow:

$$\bar{\boldsymbol{X}} = \frac{\boldsymbol{1}}{\boldsymbol{n}}(\boldsymbol{X_1} + \boldsymbol{X_2} + \cdots + \boldsymbol{X_n}) = \frac{\sum_{i/1}^{i/n} \boldsymbol{X_i}}{\boldsymbol{n}}$$

Estimator for the mean

Parameter: $\mu = E[x] = \langle x \rangle = \int_{-\infty}^{\infty} x f(x)\, dx$

Estimator: $\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i \equiv \overline{x}$ ('sample mean')

We find: $b = E[\hat{\mu}] - \mu = 0$

$$V[\hat{\mu}] = \frac{\sigma^2}{n} \qquad \left(\sigma_{\hat{\mu}} = \frac{\sigma}{\sqrt{n}} \right)$$

Sampling dist. of means

- **Theorem 3**. If the population is not infinite (of size N) or is the sampling is done without replacement, then the variance should be evaluated using:

$$\sigma'^2_{\bar{X}} = \frac{1}{n}\sigma^2\left(\frac{N-n}{N-1}\right), N \to \infty: \sigma'^2_{\bar{X}} \to \sigma^2_{\bar{X}}$$

- **Theorem 4**. If the population from which we draw samples is normally distributed with mean μ and variance σ^2, then the sample mean is also normally distributed with mean $\boldsymbol{\mu}$ and variance $\frac{\boldsymbol{\sigma^2}}{\boldsymbol{n}}$ (standard error of a mean).
- **Theorem 5**. Let's assume that the population from which samples are drawn has mean μ and variance σ^2. The population **may or may not be normally distributed**. The standardised variable associated with $\bar{X}$ can be written as:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Sampling dist. of means

- **Theorem 1**. The mean of the sample means is a consistent etimator of μ:

$$\boxed{E[\bar{X}] = \mu_{\bar{X}} = \mu}$$

where μ is the mean of the population. So, we say, that the expected value of the sample mean is the population mean – **how interesting**!

- **Theorem 2**. If a population is infinite and the sampling is random, or if a population is finite and sampling is with replacement, then the variance of the distributions of the sample means, denoted by $\sigma_{\bar{X}}$, is:

$$\boxed{E[(\bar{X} - \mu)^2] = \sigma_{\bar{X}}^2 = \frac{1}{n}\sigma^2}$$

Estimator for sample variance

- If $\{X_1, X_2, \cdots X_n\}$ denote R.Vs for a random sample of size n, the R.V. giving the variance of the sample (the sample variance) is defined as:

$$S^2 = \frac{\mathbf{1}}{\boldsymbol{n}}[(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \cdots + (X_n - \bar{X})^2]$$

- We already know, that $E[\bar{X}] = \mu$, is this the same for $E[S^2] = \sigma^2$?
  - ✓ A little digression – whenever the expected value of a statistics **is equal** to the corresponding **population parameter**, we call this statistics **an unbiased estimator**. Its value is then an unbiased estimate of the respective parameter
- Unfortunately, it can be proved that for the sample variance, we have:

$$E[S^2] = \mu_{S^2} = \frac{n-1}{n}\sigma^2$$

- However, an unbiased variance estimator is easy to find:

$$\boxed{\hat{S}^2 = \frac{n}{n-1}S^2 = \frac{\mathbf{1}}{\boldsymbol{n}-\mathbf{1}}[(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \cdots + (X_n - \bar{X})^2]}$$

$$E[\hat{S}^2] = \sigma^2$$

Estimator for the variance

Parameter: $\sigma^2 = V[x] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx$

Estimator: $\widehat{\sigma^2} = \frac{1}{\boxed{n-1}} \sum_{i=1}^{n} (x_i - \overline{x})^2 \equiv s^2$ ('sample variance')

We find:

$b = E[\widehat{\sigma^2}] - \sigma^2 = 0$ (factor of *n*-1 makes this so)

Point estimators - summary

- Sample mean $\bar{X}$ is the point estimator of parameter μ:

$$\boxed{\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n}\sum_{i=1..n} X_i}$$

- The unbiased estimator for variance is:

$$\boxed{\hat{S}^2 = \frac{1}{n-1}[(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \cdots + (X_n - \bar{X})^2] = \frac{1}{n-1}\sum (X_i - \bar{X})^2}$$

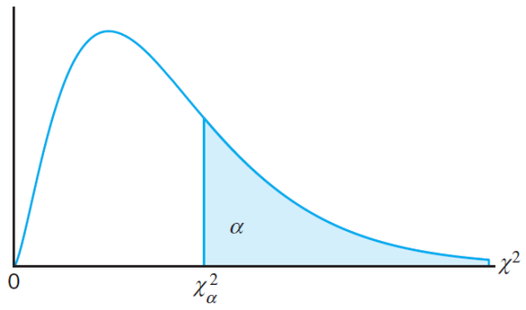
- The estimator of the correlation (X, Y) is:

$$\boxed{r(X,Y) = \frac{S_{XY}}{\sqrt{S_{XX}}\sqrt{S_{yy}}}}$$

$$S_{XX} = \sum (X_i - \bar{X})^2 \qquad S_{YY} = \sum (Y_i - \bar{Y})^2$$

$$S_{XY} = \sum (X_i - \bar{X})(Y_i - \bar{Y})$$

Sampling dist. of variances



- Sampling distributions of important statistics allow us to learn information about parameters.
  - ✓ if an engineer is interested in the population mean resistance of a certain type of resistor, the sampling distribution of $\bar{X}$ will be exploited once the sample information is gathered.
  - ✓ if the variability in resistance is to be studied, clearly the sampling distribution of S^2 will be used in learning about the parametric counterpart, the population variance σ^2.

- In order to create the **sampling distribution of variances**, we take all the possible samples of size n, that can be drawn from a population and calculate their variances.

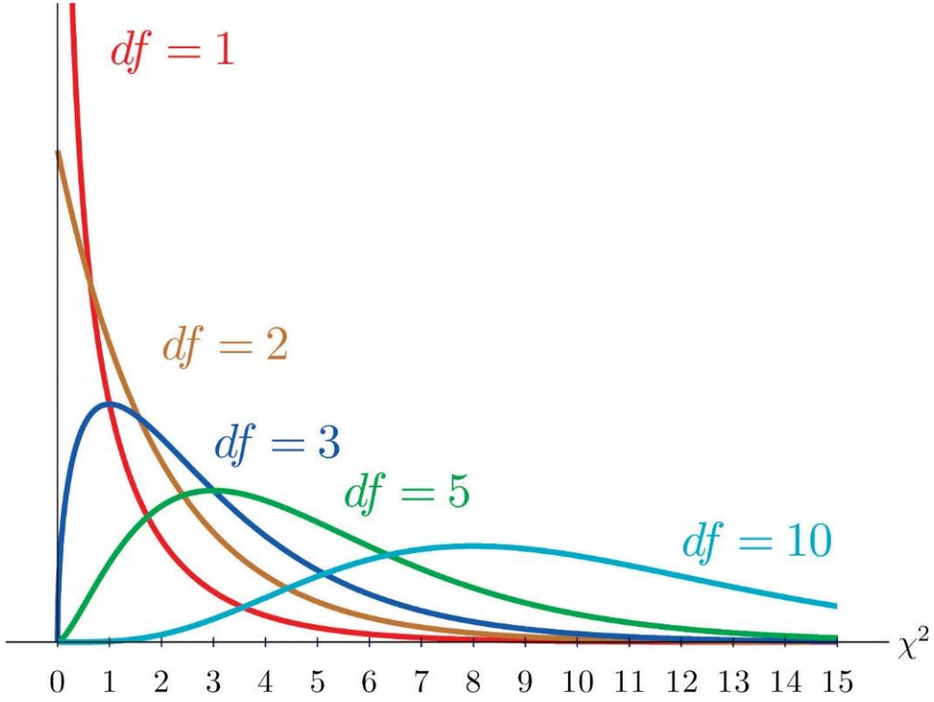
- One change is, that instead of looking directly at the distribution of the sample variance, we look at the R.V., called χ^2:

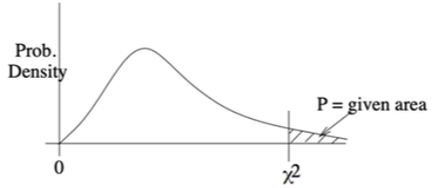
$$\chi^2 = \frac{nS^2}{\sigma^2} = \frac{(n-1)\hat{S}^2}{\sigma^2} = \frac{(X_1-\bar{X})^2 + (X_2-\bar{X})^2 + \cdots + (X_n-\bar{X})^2}{\sigma^2}$$

$$\boxed{\chi^2 = \sum_i^n \frac{(X_i-\bar{X})^2}{\sigma^2}}$$

- **Theorem 6**. If a random samples of size n are taken from a population having a normal distribution, than the sampling variable $\frac{nS^2}{\sigma^2}$ has a χ^2 distribution with $n-1$ degrees of freedom.

χ^2 distribution





	α									
d.f.	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	--	--	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.299
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997

- This is another very popular distribution in Statistics!
- The mathematical formula describing it is quite complex, again we are going to use tabulated values when solving problems!

Sampling dist. of variances

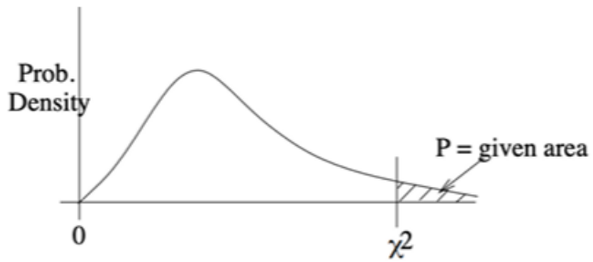
Ex. A manufacturer of car batteries guarantees that the batteries will last, on average, 3 years with a standard deviation of 1 year.

If five of these batteries have lifetimes of 1.9, 2.4, 3.0, 3.5, and 4.2 years, should the manufacturer still be convinced that the batteries have a standard deviation of **1 year**?

Assume that the battery lifetime follows a normal distribution.

$$\hat{S}^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2 = \cdots = 0.815$$

$$\chi^2 = \frac{(n-1)\hat{S}^2}{\sigma^2} = \frac{4 \cdot 0.815}{1} = 3.26$$



d.f.	α									
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	--	--	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860

95%

$\chi^2 = 3.26$ is between $\chi^2_{0.95}$ and $\chi^2_{0.05}$,
the manufactures has no reason to suspect that the standard deviation is other than 1 year.

The method of moments (MoM)

- We use the common sense when introducing this approach
- We introduced the notions of population moments and sample moments, the former are **unknown** and the latter are calculated using **data samples.**
- The **method of moments** (MoM) uses the **sample moments** and match them to the analogous **population moments** to obtain **estimates for the unknown parameters**. Seems simple…

Population moments:

$$\mu_r = E[X^r]$$

Sample moments – functions of the parameters θ_i:

$$m_r = \frac{1}{n}\sum X_i^r$$

one need to solve a system of equations for θ_i

- The Method of Moments replaces **expectations with averages.**
- Simple but not always optimal, often used as a first aproximation.

The method of moments (MoM)

- Let's write down explicitly moment estimators for a few most popular models we discussed so far.
- The **exponential** $f(x;\lambda) = \lambda e^{-\lambda x}$: one unknown parameter – the mean of the distribution. Matching sample and population moments gives the following estimate:

$$\left.\begin{aligned} \mu_1 &= E[X] = 1/\lambda \\ m_1 &= \bar{X} = 1/\lambda \end{aligned}\right\} \quad \hat{\lambda} = \frac{1}{\bar{X}}$$

- The Bernouli distribution $\mathcal{B}(m,p)$: we have one unknown parameter p. The matching procedure gives:

$$\mu_1 = E[X] = p \quad \text{and} \quad p = \frac{1}{n}\sum X_i \quad \text{so:} \quad \hat{p} = \frac{1}{n}\sum X_i$$

- A normal distribution $\mathcal{N}(\mu,\sigma^2)$, we use sample mean and variance:

$$\left.\begin{aligned} \mu_1 &= E[X] = \mu & E[X^2] &= \mu^2 + \sigma^2 \\ m_1 &= \bar{X} = \mu & m_2 &= \tfrac{1}{n}\textstyle\sum X_i^2 \end{aligned}\right\} \quad \begin{aligned} \hat{\mu} &= \tfrac{1}{n}\textstyle\sum X_i \\ \hat{\sigma}^2 &= \frac{1}{n}\sum X_i^2 - \mu^2 = \frac{1}{n}\sum X_i^2 - \bar{X}^2 = \frac{1}{n}\sum (X_i - \bar{X})^2 \equiv s^2 \end{aligned}$$

The method of moments (MoM)

- In practice, we are going to observe some fluctuations, let's consider the following: we used a generator of random numbers distributed according to the Poisson model. We draw two samples:

$$X_{(1)} = \{5,5,2,3,4,6,4,1\}, X_{(2)} = \{4,2,5,2,4,1,1,1\}$$

$$\hat{\mu} = \bar{x}_{(1)} = \frac{30}{8} = 3.75$$

$$\hat{\mu} = \bar{x}_{(2)} = \frac{20}{8} = 2.50$$

$$P(X = k) = \frac{e^{-\lambda}\,\lambda^k}{k!}$$

$$m_1 = \bar{X} = \lambda$$

- These variations are „normal", we are going to observe them

- The point is that (SV – sample variance):

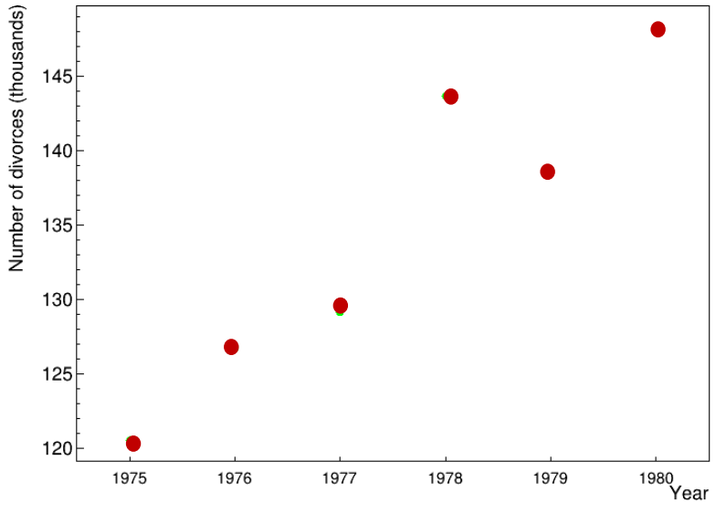
$$E[\bar{X}] = \mu, \qquad SV[\bar{X}] = \sqrt{\frac{\sigma^2}{n}} = \frac{\lambda}{n}$$

As the sample size n increases, the estimator λ converges to the true value of λ

More than one way…

- Lets inspect the following data regarding the number of divorces in different years in some country in Europe

Year	1975	1976	1977	1978	1979	1980
# divorces (10^3)	120.5	126.7	129.1	143.7	138.7	148.3



- Interesting…, very tempting to fit a model right away.

More than one way…

- From the plot we could conclude, that the **true underlaying distribution** describing the data can be represented by a **linear model**
- From the data we also conclude that **the slope** of the line is positive – ok, the task is then to **estimate this slope**, α, and then we could predict the annual rate of increase of divorces
- But how do we do that? <u>It is not so obvious like the water example</u>(??)
- Consider this:
  - $\hat{\alpha}_1$- join the first and the last point
  - $\hat{\alpha}_2$ - join the mid-points P_1P_2 and P_5P_6
  - $\hat{\alpha}_3$ - join the centroid of the first triplet and the second one
- Mind you, these are all sensible options!

The best parameters

- Let's do the calculations explicitly

$$\hat{\alpha}_1 = \frac{148.3 - 120.5}{80 - 75} = \frac{27.8}{5} = 5.6$$

$$\hat{\alpha}_2 = \frac{(138.7 + 148.3)/2 - (120.5 + 126.7)/2}{79.5 - 75.5} = \frac{19.9}{4} = 5.0$$

$$\hat{\alpha}_3 = \frac{(143.7 + 138.7 + 148.3)/3 - \cdots}{79 - 76} = \frac{18.13}{3} = 6.0$$

- So, is there a way to make a judgement on **which one of these estimates is the „best"**? **And what exactly the best means**?
- The second question actually pertains to **the estimator properties** and not the estimate (a number we obtained)
- So, we need to look at the properties of the **sampling distribution** of respective estimators!

Analysis

- Mind one thing. This example is not quite what we could call an experiment – we cannot repeat year 1976 and check the number of divorces again…
- However, we can still evaluate the deviations of data from the predicted model and treat them as random variable
- Say, the difference (**residual**) r is defined as follow:

$$r_i = y_i - (\beta + \alpha x_i)$$

- Next, we assume that the residual (random variable) will have a mean value and variance: μ_r and σ_r^2
- Further, we assume that <u>the residual should have mean value equal to 0</u> (think about that!), so finally our model for the given data set is:

$$Y_i = \beta + \alpha x_i + r_i$$

- We have therefore three parameters $(\alpha, \beta, \sigma_r^2)$

Analysis

- Having formulated the model we can now start discussing the properties of the sampling distributions of our estimators
- So, we are going to treat **the estimate** (a number evaluated using the data):

$$\hat{\alpha}_1 = \frac{y_6 - y_1}{x_6 - x_1}$$

- … as a single measurement (observation) of the random variable (the estimator)

$$\hat{\alpha}_1 = \frac{Y_6 - Y_1}{x_6 - x_1}$$

- To come up with the answer regarding how good is such estimator we start from working out its mean and variance (we use the knowledge of these function of R.V. remembering that α, β, x_i are just constant numbers)

$$E[Y_i] = \beta + \alpha x_i$$

$$V[Y_i] = \sigma_r^2$$

$$E[r_i] = 0, V[r_i] = \sigma_r^2$$

Analysis

- Now, we can calculate the expected value of $\hat{\alpha}_1$:

$$E[\hat{\alpha}_1] = E\left[\frac{Y_6 - Y_1}{x_6 - x_1}\right] = \frac{1}{x_6 - x_1} E[Y_6 - Y_1] =$$

$$= \frac{1}{x_6 - x_1}(E[Y_6] - E[Y_1]) = \frac{1}{x_6 - x_1}\big((\beta + \alpha x_6) - (\beta + \alpha x_1)\big)$$

$$E[\hat{\alpha}_1] = \frac{1}{x_6 - x_1}(\alpha x_6 - \alpha x_1) = \alpha$$

- Neat! The expected value of the estimator is exactly equal to the unknown parameter. Good job
- What about the other estimators?

$$\hat{\alpha}_2 = \frac{\frac{1}{2}(Y_5 + Y_6) - \frac{1}{2}(Y_1 + Y_2)}{\frac{1}{2}(x_5 + x_6) - \frac{1}{2}(x_1 + x_2)}$$

$$\hat{\alpha}_3 = ?$$

Analysis

- Repeating the same calculations for remaining two estimators we conclude that their expected values are always the same and equal exactly the unknown parameter we estimating
- In general we say that an estimator $\hat{\theta}$, which we use to estimate an unknown parameter of a model, is **unbiased for parameter $\boldsymbol{\theta}$** if the following is true:

$$E\left[\hat{\theta}\right] = \theta$$

- So, it seems that all of them doing just fine. What next we can check…? The variance!
- In this case the best option would be to choose the one that features the least variability about its mean value, so:

$$V[\hat{\alpha}_1] = V\left[\frac{Y_6 - Y_1}{x_6 - x_1}\right] = \frac{1}{(x_6 - x_1)^2}(V[Y_6] + V[-Y_1]) =$$

$$= \frac{1}{(x_6 - x_1)^2}(\sigma_r^2 + \sigma_r^2) = \frac{2\sigma_r^2}{(x_6 - x_1)^2} = \frac{2\sigma_r^2}{25}$$

Analysis

- Again, we can repeat the calculations for the remaining two estimators (you are encuraged to do so!)
- We get the following:

$$V[\hat{\alpha}_2] = \frac{4\sigma_r^2}{64} \qquad V[\hat{\alpha}_3] = \frac{6\sigma_r^2}{81}$$

- So: $V[\hat{\alpha}_2] < V[\hat{\alpha}_3] < V[\hat{\alpha}_1]$
- Using the variance we say that the best (most efficient) estimator is the $\hat{\alpha}_2$ - thus we have the winner!

Summary so far

- A generic „algorithm” for point estimation task would be:

  - ✓ **Collect the data** and understand it
  - ✓ Come up with a **model**, this will specify a **parameter** of many parameters that we need to estimate
  - ✓ For a given parameter we need an **estimator** (typically we will concentrate on the mean value or variance, however we also can tackle more ambitious cases – divorces)
  - ✓ Work out the **estimate of the parameter** – this is a random variable and will be different for different data sets
  - ✓ Finally, analyse the **sampling distribution of the estimator** to make a judgement of its usefulness
  - ✓ We are looking for **unbiased** (expectation value) and **efficient** estimators (variance)

Can we do better?

- That was fun! And we learned a lot, however following such generic path each time we need to make an estimate **seems too much**
- We need a technique(s) that allows us **to define sensible estimators** (again, we could spend a lifetime on deriving estimators that are reasonable)
- So, such a technique would „automatically” **come up with a formula for best estimators**

- One thing to remember – there is no universal method to achieve the above task, in time a number of approaches have been proposed. <u>There is no „best” one</u>
- We concentrate on three techniques: the method **of least squares**, the method of **moments** and the method of **maximum likelihood**

The method of least squares (MLS)

- Say, we are interested in estimating the mean value of some distribution which is θ. We take data sample: $\{X_1, X_2, \dots, X_n\}$
- Since the mean is „a typical" value, we conclude that the respective differences $X_i - \theta$ should be „small" (simultaneously)
- Also, the sum of squares of these differences should be as small as possible:

$$\boxed{S = \sum_i (X_i - \theta)^2 \to min}$$

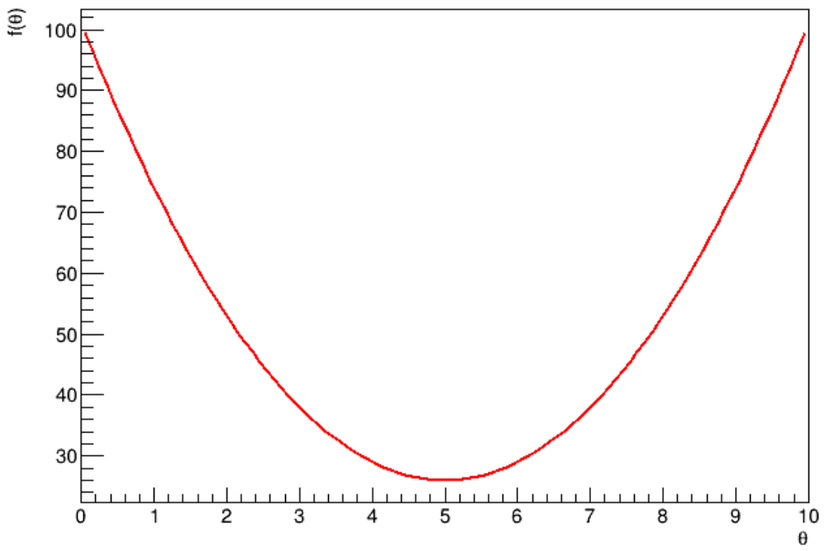
- This is the method of least squares (MLS)
- As usual, an **example** is in order! Say we collected sample: $X = \{2,4,9\}$ and we want to estimate the mean value of the parent distribution these numbers came from. Applying the MLS

$$S_3 = \sum_{i/1}^{i/3} (X_i - \theta)^2 = (2-\theta)^2 + (4-\theta)^2 + (9-\theta)^2 = \cdots =$$

$$= 101 - 30\theta + 3\theta^2$$

The method of least squares (MLS)

- This function will take different values for different θ – task is to identify the θ for which the sum is the smallest



- The minimum is attained at the point $\theta = 5$. We can also check that the sample mean, $\bar{x} = 5$. Reasonable!
- So, the method gives results compatible with the common sense, that is encouraging!

The method of least squares (MLS)

- In a general case where we have a random sample of size n drawn from a population with an unknown mean value:

$$S_n = \sum_{i/1}^{i/n} (X_i - \theta)^2 = \sum_{i/1}^{i/n} \left(X_i^2 - 2\theta X_i + \theta^2\right) =$$

$$= \sum_{i/1}^{i/n} X_i^2 + 2\theta \sum_{i/1}^{i/n} X_i + n\theta^2$$

- This formula is minimised by: $\theta = \frac{1}{n}\sum_{i/1}^{i/n} X_i = \bar{X}$ - the sample mean
- So, the sample mean $\bar{x}$ of a random sample X taken from a population of an unknown mean value (we call is here in a generic way θ) is the **least squares estimator** $\hat{\theta}$ (or $\hat{\mu}$)
- The language here is important, so we are precise about what we mean!

we will continue the discussion how to obtain the best estimators in a few weeks!